

Multimodal Fashion & Context Retrieval System

End-to-End System Design & Architecture

1. Introduction

The objective of this project is to build an intelligent multimodal image retrieval system capable of understanding **what a person is wearing, where they are, and the overall style or vibe** of the image from natural language queries.

Unlike traditional image search systems that rely solely on global image embeddings, this project combines multiple complementary visual understanding modules to improve retrieval accuracy for fashion-specific queries.

The system is designed specifically for compositional fashion retrieval where a user may search for complex descriptions such as:

- "A woman wearing a white blazer inside an office."
- "Someone in a blue hoodie walking in a park."
- "A red tie with a white shirt in a formal setting."
- "Casual weekend outfit for a city walk."

These queries require understanding multiple visual concepts simultaneously:

- Clothing items
- Clothing colors
- Human style
- Environment
- Semantic similarity

The proposed architecture combines all these signals into one hybrid retrieval pipeline.

2. Problem Statement

Traditional CLIP-based retrieval systems generate a single embedding for the entire image.

Although this works well for generic semantic retrieval, it struggles with fashion-specific compositional queries.

For example,

Query:

> A red tie with a white shirt inside an office.

A vanilla CLIP system may retrieve

- White tie with red shirt
- Red office furniture
- White shirts without ties

because all visual concepts are compressed into a single embedding.

The system lacks explicit understanding of

- which object is red
- which clothing item is white
- where the scene is located
- whether the outfit is formal or casual

To overcome these limitations, our architecture introduces multiple specialized understanding modules that work together during retrieval.

3. Design Goals

The system is designed around the following objectives.

Understand Clothing

Identify clothing categories such as

- Shirt
- T-Shirt
- Hoodie

- Coat
 - Blazer
 - Dress
 - Tie
 - Pants
 - Shoes
 - Accessories
-

Understand Colors

Identify dominant clothing colors including

- Red
 - Blue
 - Yellow
 - Black
 - White
 - Brown
 - Green
-

Understand Scene

Recognize environmental context such as

- Office
 - Home
 - Street
 - Park
 - Mall
 - Café
-

Understand Style

Infer overall fashion style

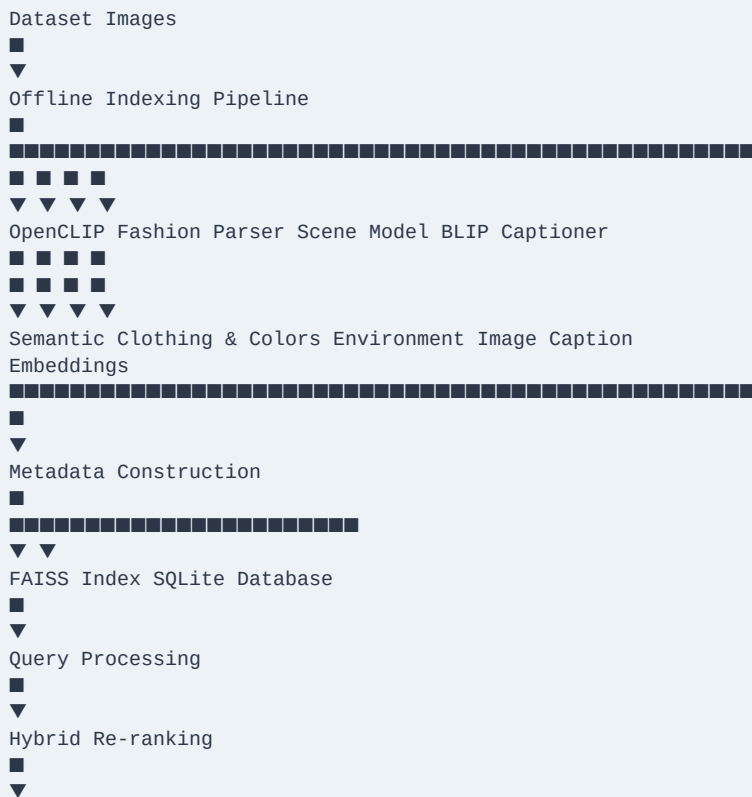
Examples

- Formal
- Casual
- Business
- Streetwear
- Weekend
- Smart Casual

Preserve Semantic Understanding

Maintain CLIP's strong zero-shot semantic capability for natural language retrieval.

4. High-Level Architecture



5. Complete Pipeline

The project consists of two major stages.

Stage 1

Offline Image Indexing

Stage 2

Online Query Retrieval

6. Offline Image Indexing

The indexing stage processes every image once and stores all extracted information.

This stage consists of six modules.

Module 1 — OpenCLIP Feature Extraction

Purpose

Generate a global semantic embedding for every image.

Input

Image

Output

512/768 dimensional embedding depending on CLIP backbone.

This embedding captures

- Objects
- Overall appearance
- Scene semantics
- Human pose
- Clothing semantics

These embeddings are stored inside FAISS for fast nearest-neighbor search.

Module 2 — Fashion Attribute Extraction

Purpose

Extract structured clothing information.

Possible outputs

Clothing Categories

- Shirt
- Hoodie
- Blazer
- Coat
- Pants
- Dress
- Skirt
- Tie
- Shoes
- Bag

Color Information

- Shirt Color

- Pant Color
- Dress Color
- Shoe Color

Style Information

- Formal
- Casual
- Streetwear
- Business

This information becomes structured metadata.

Module 3 — Scene Classification

Purpose

Recognize where the person is located.

Possible scene labels

- Office
- Park
- Home
- Street
- Café
- Shopping Mall

This solves contextual queries.

Example

Professional business attire inside a modern office.

Module 4 — Image Caption Generation

Purpose

Generate a natural language description of every image.

Example

"A woman wearing a blue blazer sitting inside an office."

The generated caption helps bridge the gap between visual understanding and language understanding.

Module 5 — Metadata Builder

The outputs of all previous modules are merged into one metadata object.

Example

```
{
  "id": 101,
  "caption": "A man wearing a white shirt and red tie inside an office.",
  "scene": "office",
  "style": "formal",
  "clothes": [
    "shirt",
    "tie",
    "pants"
  ],
  "colors": {
    "shirt": "white",
    "tie": "red",
    "pants": "black"
  }
}
```

This metadata is stored inside SQLite.

Module 6 — Storage

Two storage systems are maintained.

FAISS

Stores

- CLIP embeddings

Purpose

Fast semantic retrieval.

SQLite

Stores

- captions
- colors
- clothing
- style
- scene
- image path

Purpose

Efficient metadata filtering and reranking.

7. Online Query Retrieval

When the user enters a query, the retrieval pipeline begins.

Example

> Someone wearing a blue shirt sitting on a park bench.

Step 1 — Query Parsing

The query is decomposed into structured attributes.

Extracted components

Scene

Park

Objects

Bench

Clothing

Shirt

Color

Blue

Activity

Sitting

This structured representation enables fine-grained matching.

Step 2 — Query Embedding

The complete natural language query is encoded using OpenCLIP.

This embedding is used for semantic retrieval from FAISS.

Step 3 — Candidate Retrieval

FAISS returns the Top-N semantically similar images.

Example

Top 100 candidates.

Step 4 — Metadata Matching

Each retrieved candidate is evaluated using structured metadata.

Checks include

- Clothing category match
 - Clothing color match
 - Scene match
 - Style match
 - Caption similarity
-

Step 5 — Hybrid Re-ranking

Instead of relying only on CLIP similarity, the final ranking combines multiple scores.

Example scoring function

Final Score

=

$0.45 \times \text{CLIP Similarity}$

+

$0.20 \times \text{Fashion Attribute Match}$

+

$0.15 \times \text{Scene Match}$

+

$0.10 \times \text{Caption Similarity}$

+

0.10 × Style Match

This significantly improves precision on fashion-based queries.

8. Why Hybrid Retrieval?

Different modules specialize in different visual concepts.

| Module | Responsibility |

|-----|-----|

| OpenCLIP | Global semantic understanding |

| Fashion Parser | Clothing & colors |

| Scene Classifier | Environmental context |

| BLIP Caption | Language alignment |

| SQLite Metadata | Structured filtering |

| FAISS | Fast vector retrieval |

Instead of depending on one model to understand everything, each model contributes its own expertise.

9. Scalability

The retrieval process remains efficient even for very large datasets.

For one million images

Images

↓

OpenCLIP Embeddings



FAISS IVF Index



Top 200 Candidates



Metadata Filtering



Hybrid Re-ranking



Top 10 Results

Only the candidate images undergo reranking, making the system highly scalable.

10. Advantages

- ✓ Better than vanilla CLIP
- ✓ Understands clothing attributes
- ✓ Understands clothing colors
- ✓ Understands environmental context
- ✓ Supports compositional queries
- ✓ Zero-shot retrieval capability
- ✓ Modular architecture

✓ Easily extendable

✓ Scalable to millions of images

11. Future Improvements

Geographic Understanding

Integrate GeoCLIP to recognize

- Cities
 - Tourist locations
 - Famous landmarks
-

Weather Understanding

Recognize

- Rain
- Snow
- Sunny
- Cloudy
- Fog

This enables queries like

> A person wearing a raincoat on a rainy street.

Improved Fashion Parsing

Fine-tune FashionCLIP or train on DeepFashion2/Fashionpedia for better clothing recognition.

Learning-Based Re-ranking

Replace manually weighted fusion with a trainable cross-encoder that learns optimal feature importance.

User Feedback Loop

Allow user feedback to improve ranking over time through relevance learning.

12. Conclusion

This project presents a modular multimodal retrieval architecture designed specifically for fashion-aware image search.

Rather than relying solely on global image embeddings, the system combines semantic embeddings, structured fashion attributes, scene understanding, caption generation, and metadata-aware reranking into a unified retrieval pipeline.

The hybrid design improves compositional reasoning, contextual understanding, and fashion-specific retrieval while remaining scalable to datasets containing millions of images.

The architecture is modular, extensible, and well suited for real-world multimodal fashion search applications.